Supplementary Material

We provide additional supplementary material in this Appendix. We refer to Table 3 for an overview of Friedman tests comparing the responses across the four exposure phases. Similarly, we refer to Table 4 for an overview of Friedman tests comparing the responses across the four stimuli. Table 5 reports additional post-hoc comparisons for statistically significant tests in Table 3. Likewise, we present post-hoc comparisons for Table 4 in Table 6. Lastly, the non-normality plots of all measured features are presented in Table 7.

		Phase				Statistics			
Measurement	Stimulus	pre	first	post	second	n	χ^2	p	p_{FDR}
Head elevation	social°	-1.37	-27.18	-18.86	-23.63	9	10.33	0.016	0.029*
	social$^+$	-3.58	-36.93	-22.68	-44.60	9	4.60	0.204	0.278
	food°	2.83	15.12	-11.52	5.16	9	8.73	0.033	0.056
	food$^+$	0.94	6.52	-11.85	8.29	9	9.80	0.020	0.036*
Cheek caudoventral	social°	-0.44	-1.52	-2.17	-1.80	9	13.40	0.004	0.011*
	social$^+$	0.13	-0.99	-0.98	-1.93	9	5.53	0.137	0.196
	food°	-0.15	-0.15	-1.48	-0.93	9	15.80	0.001	0.005**
	food$^+$	-0.22	-0.26	-1.71	-0.46	9	12.47	0.006	0.015*
Nose: Pull towards glabella	social°	-0.13	0.00	-0.14	0.03	9	8.33	0.040	0.063
	social$^+$	0.08	0.28	0.06	0.13	9	6.73	0.081	0.119
	food°	-0.04	-0.13	-0.15	-0.22	9	4.87	0.182	0.255
	food$^+$	-0.00	-0.10	-0.15	-0.22	9	3.80	0.284	0.370
Below eye: dorsal	social°	0.32	0.39	0.68	0.51	9	11.00	0.012	0.024*
	social$^+$	-0.23	0.44	0.84	1.04	9	11.00	0.012	0.024*
	food°	-0.33	-0.09	0.64	0.24	9	13.53	0.004	0.011*
	food$^+$	-0.15	0.05	0.61	0.29	9	16.07	0.001	0.005**
Eyebrow: caudal	social°	0.02	-0.00	-0.09	-0.05	9	3.40	0.334	0.425
	social$^+$	0.05	0.09	-0.01	-0.04	9	2.47	0.481	0.550
	food°	-0.01	-0.21	-0.17	-0.37	9	12.07	0.007	0.017*
	food$^+$	0.04	-0.15	-0.16	-0.25	9	7.13	0.068	0.105
Eyebrow: medial	social°	0.04	-0.17	0.05	-0.17	9	10.47	0.015	0.028*
	social$^+$	-0.01	-0.28	-0.07	-0.27	9	11.53	0.009	0.021*
	food°	-0.05	0.00	0.14	-0.11	9	7.00	0.072	0.109
	food$^+$	-0.00	-0.01	0.12	0.02	9	10.73	0.013	0.026*
Forehead: caudal	social°	-0.12	0.29	-0.20	0.24	9	15.00	0.002	0.006**
	social$^+$	0.16	0.62	0.15	0.57	9	13.53	0.004	0.011*
	food°	0.03	-0.34	-0.44	-0.54	9	20.07	<0.001	0.001**
	food$^+$	-0.10	-0.40	-0.53	-0.63	9	11.27	0.010	0.023*
Forehead: medial	social°	0.11	-1.13	-0.35	-1.11	9	17.93	<0.001	0.002**
	social$^+$	0.01	-1.62	-0.65	-1.54	9	22.60	<0.001	<0.001***
	food°	-0.09	-0.13	-0.01	-0.07	9	3.93	0.269	0.358
	food$^+$	-0.07	-0.22	0.02	-0.17	9	12.60	0.006	0.015*
Ear 1: dorsal	social°	0.43	-3.70	-1.08	-3.60	9	22.73	<0.001	<0.001***
	social$^+$	-0.01	-5.25	-2.07	-4.72	9	22.73	<0.001	<0.001***
	food°	-0.34	-0.51	-0.19	-0.33	9	2.73	0.435	0.518
	food$^+$	0.14	-0.40	0.05	-0.33	9	8.60	0.035	0.058
Ear 2: medial	social°	0.40	-2.94	-2.19	-3.04	9	20.33	<0.001	0.001**
	social$^+$	0.39	-3.61	-2.56	-3.94	9	23.13	<0.001	<0.001***
	food°	0.02	-1.15	-1.12	-1.38	9	15.80	0.001	0.005**
	food$^+$	-0.13	-1.29	-1.09	-1.52	9	21.13	<0.001	0.001**
Ear 5: medial	social°	0.26	-2.36	-0.66	-2.30	9	13.93	0.003	0.010**
	social$^+$	-0.23	-3.74	-1.63	-3.64	9	18.87	<0.001	0.002**
	food°	-0.48	-0.07	-0.06	0.09	9	1.93	0.586	0.631
	food$^+$	0.09	0.16	0.26	0.19	9	1.80	0.615	0.650
Tail: wag amplitude	social°	1.46	0.02	-5.46	-2.09	9	16.73	<0.001	0.004**
	social$^+$	0.87	4.16	-4.57	0.84	9	10.73	0.013	0.026*
	food°	0.57	-5.17	-8.12	-9.51	9	17.80	<0.001	0.002**
	food$^+$	3.71	-2.01	-6.19	-7.68	9	18.73	<0.001	0.002**

Table 3: Friedman tests comparing the responses across the four exposure phases, computed separately for each measurement and stimulus. We first average each dog's values across all four recording sessions within an exposure phase, resulting in one value per dog and phase. The phase columns report the mean of the per-dog values as deviations from their respective pre-exposure baseline (averaged over all stimuli). Deviations within on measurement across the pre-exposure phase sum to zero by construction. A significant χ^2 indicates that the measurement differs between phases. p_{FDR} reports the Benjamini-Hochberg adjusted p-values, corrected once across all measurements and stimuli.

		Stimulus				Statistics			
Measurement	**Phase**	social°	social^{+}	food°	food^{+}	n	χ^2	p	p_{FDR}
Head elevation	pre	-1.37	-3.58	2.83	0.94	9	5.53	0.137	0.219
	first	-27.18	-36.93	15.12	6.52	9	17.67	<0.001	0.003**
	post	-18.86	-22.68	-11.52	-11.85	9	2.47	0.481	0.627
	second	-23.63	-44.60	5.16	8.29	9	14.73	0.002	0.009**
Cheek caudoventral	pre	-0.44	0.13	-0.15	-0.22	9	0.60	0.896	0.930
	first	-1.52	-0.99	-0.15	-0.26	9	1.13	0.769	0.861
	post	-2.17	-0.98	-1.48	-1.71	9	3.80	0.284	0.397
	second	-1.80	-1.93	-0.93	-0.46	9	3.93	0.269	0.386
Nose: Pull towards glabella	pre	-0.13	0.08	-0.04	-0.00	9	11.00	0.012	0.030*
	first	0.00	0.28	-0.13	-0.10	9	9.40	0.024	0.053
	post	-0.14	0.06	-0.15	-0.15	9	10.73	0.013	0.031*
	second	0.03	0.13	-0.22	-0.22	9	3.53	0.316	0.432
Below eye: dorsal	pre	0.32	-0.23	-0.33	-0.15	9	0.73	0.865	0.930
	first	0.39	0.44	-0.09	0.05	9	11.93	0.008	0.022*
	post	0.68	0.84	0.64	0.61	9	0.20	0.978	0.978
	second	0.51	1.04	0.24	0.29	9	9.13	0.028	0.057
Eyebrow: caudal	pre	0.02	0.05	-0.01	0.04	9	1.27	0.737	0.860
	first	-0.00	0.09	-0.21	-0.15	9	6.07	0.108	0.184
	post	-0.09	-0.01	-0.17	-0.16	9	0.60	0.896	0.930
	second	-0.05	-0.04	-0.37	-0.25	9	4.07	0.254	0.375
Eyebrow: medial	pre	0.04	-0.01	-0.05	-0.00	9	1.40	0.706	0.841
	first	-0.17	-0.28	0.00	-0.01	9	8.33	0.040	0.076
	post	0.05	-0.07	0.14	0.12	9	13.13	0.004	0.014*
	second	-0.17	-0.27	-0.11	0.02	9	12.33	0.006	0.020*
Forehead: caudal	pre	-0.12	0.16	0.03	-0.10	9	6.73	0.081	0.142
	first	0.29	0.62	-0.34	-0.40	9	15.00	0.002	0.008**
	post	-0.20	0.15	-0.44	-0.53	9	9.80	0.020	0.046*
	second	0.24	0.57	-0.54	-0.63	9	14.20	0.003	0.011*
Forehead: medial	pre	0.11	0.01	-0.09	-0.07	9	7.53	0.057	0.106
	first	-1.13	-1.62	-0.13	-0.22	9	19.00	<0.001	0.002**
	post	-0.35	-0.65	-0.01	0.02	9	20.07	<0.001	0.002**
	second	-1.11	-1.54	-0.07	-0.17	9	18.73	<0.001	0.002**
Ear 1: dorsal	pre	0.43	-0.01	-0.34	0.14	9	5.93	0.115	0.189
	first	-3.70	-5.25	-0.51	-0.40	9	17.40	<0.001	0.003**
	post	-1.08	-2.07	-0.19	0.05	9	19.80	<0.001	0.002**
	second	-3.60	-4.72	-0.33	-0.33	9	20.47	<0.001	0.002**
Ear 2: medial	pre	0.40	0.39	0.02	-0.13	9	8.33	0.040	0.076
	first	-2.94	-3.61	-1.15	-1.29	9	13.40	0.004	0.013*
	post	-2.19	-2.56	-1.12	-1.09	9	19.93	<0.001	0.002**
	second	-3.04	-3.94	-1.38	-1.52	9	24.73	<0.001	<0.001***
Ear 5: medial	pre	0.26	-0.23	-0.48	0.09	9	5.40	0.145	0.225
	first	-2.36	-3.74	-0.07	0.16	9	17.13	<0.001	0.003**
	post	-0.66	-1.63	-0.06	0.26	9	13.93	0.003	0.011*
	second	-2.30	-3.64	0.09	0.19	9	19.00	<0.001	0.002**
Tail: wag amplitude	pre	1.46	0.87	0.57	3.71	9	2.33	0.506	0.630
	first	0.02	4.16	-5.17	-2.01	9	11.00	0.012	0.030*
	post	-5.46	-4.57	-8.12	-6.19	9	2.33	0.506	0.630
	second	-2.09	0.84	-9.51	-7.68	9	11.27	0.010	0.029*

Table 4: Friedman tests comparing the responses across the four stimuli, computed separately for each measurement and exposure phase. We first average each dog's values across all four recording sessions within an exposure phase, resulting in one value per dog and phase. The stimuli columns report the mean of the per-dog values as deviations from their respective pre-exposure baseline (averaged over all stimuli). Deviations within on measurement across the pre-exposure phase sum to zero by construction. A significant χ^2 indicates that the measurement differs between stimuli. p_{FDR} reports the Benjamini-Hochberg adjusted p-values, corrected once across all measurements and exposure phase.

Measurement	Stimulus	pre vs first	pre vs post	pre vs second	first vs post	first vs second	post vs second
Head elevation	social°	0.031*	0.094	0.035*	0.783	0.744	0.744
	social+	--	--	--	--	--	--
	food°	--	--	--	--	--	--
	food+	0.760	0.744	0.801	0.031*	0.820	0.031*
Cheek caudoventral	social°	0.035*	0.014*	0.039*	0.451	0.684	0.498
	social+	--	--	--	--	--	--
	food°	1.000	0.014*	0.070	0.014*	0.039*	0.121
	food+	0.778	0.039*	0.734	0.014*	0.638	0.014*
Nose: Pull towards glabella	social°	--	--	--	--	--	--
	social+	--	--	--	--	--	--
	food°	--	--	--	--	--	--
	food+	--	--	--	--	--	--
Below eye: dorsal	social°	0.172	0.172	0.172	0.167	0.314	0.244
	social+	0.073	0.067	0.031*	0.359	0.031*	0.286
	food°	0.207	0.031*	0.119	0.031*	0.167	0.073
	food+	0.314	0.119	0.167	0.031*	0.172	0.031*
Eyebrow: caudal	social°	--	--	--	--	--	--
	social+	--	--	--	--	--	--
	food°	0.117	0.155	0.117	0.910	0.155	0.148
	food+	--	--	--	--	--	--
Eyebrow: medial	social°	0.134	0.855	0.134	0.042*	0.964	0.042*
	social+	0.044*	0.964	0.035*	0.044*	0.964	0.042*
	food°	--	--	--	--	--	--
	food+	1.000	0.211	0.944	0.035*	0.903	0.044*
Forehead: caudal	social°	0.023*	0.518	0.087	0.016*	0.401	0.019*
	social+	0.016*	0.734	0.518	0.033*	0.518	0.016*
	food°	0.023*	0.016*	0.016*	0.146	0.016*	0.454
	food+	0.033*	0.019*	0.019*	0.401	0.019*	0.511
Forehead: medial	social°	0.013*	0.041*	0.009**	0.013*	0.734	0.009**
	social+	0.009**	0.009**	0.009**	0.013*	0.642	0.009**
	food°	--	--	--	--	--	--
	food+	0.211	0.691	0.431	0.009**	0.211	0.009**
Ear 1: dorsal	social°	0.006**	0.006**	0.006**	0.009**	1.000	0.006**
	social+	0.006**	0.006**	0.006**	0.009**	0.392	0.006**
	food°	--	--	--	--	--	--
	food+	--	--	--	--	--	--
Ear 2: medial	social°	0.008**	0.008**	0.008**	0.094	0.518	0.018*
	social+	0.008**	0.008**	0.008**	0.039*	0.232	0.008**
	food°	0.008**	0.008**	0.013*	1.000	0.197	0.073
	food+	0.008**	0.013*	0.008**	0.392	0.018*	0.008**
Ear 5: medial	social°	0.016*	0.047*	0.016*	0.029*	0.820	0.036*
	social+	0.016*	0.016*	0.016*	0.016*	0.222	0.020*
	food°	--	--	--	--	--	--
	food+	--	--	--	--	--	--
Tail: wag amplitude	social°	0.193	0.052	0.193	0.012*	0.541	0.137
	social+	0.652	0.257	0.541	0.012*	0.193	0.109
	food°	0.012*	0.012*	0.012*	0.257	0.232	0.431
	food+	0.012*	0.012*	0.012*	0.085	0.085	0.652

Table 5: Post-hoc Wilcoxon signed-rank tests between pairs of different exposure phases, computed separately for each measurement and stimulus. We report post-hoc tests only for corresponding significant Friedman tests ($p_{FDR} \leq 0.05$, Table 3). – denotes no significant corresponding Friedman test. All reported p-values are Benjamini-Hochberg corrected within each measurement across all tested pairs.

Measurement	**Phase**	social° vs social^{+}	social° vs food°	social° vs food^{+}	social^{+} vs food°	social^{+} vs food^{+}	food° vs food^{+}
Head elevation	pre	--	--	--	--	--	--
	first	0.155	0.023*	0.067	0.023*	0.055	0.426
	post	--	--	--	--	--	--
	second	0.130	0.082	0.055	0.035*	0.035*	0.179
Cheek caudoventral	pre	--	--	--	--	--	--
	first	--	--	--	--	--	--
	post	--	--	--	--	--	--
	second	--	--	--	--	--	--
Nose: Pull towards glabella	pre	0.047*	0.281	0.219	0.258	0.595	0.568
	first	--	--	--	--	--	--
	post	0.070	0.622	0.568	0.258	0.258	0.910
	second	--	--	--	--	--	--
Below eye: dorsal	pre	--	--	--	--	--	--
	first	0.570	0.117	0.193	0.193	0.193	0.570
	post	--	--	--	--	--	--
	second	--	--	--	--	--	--
Eyebrow: caudal	pre	--	--	--	--	--	--
	first	--	--	--	--	--	--
	post	--	--	--	--	--	--
	second	--	--	--	--	--	--
Eyebrow: medial	pre	--	--	--	--	--	--
	first	--	--	--	--	--	--
	post	0.059	0.431	0.219	0.031*	0.023*	0.570
	second	0.078	0.570	0.066	0.193	0.023*	0.167
Forehead: caudal	pre	--	--	--	--	--	--
	first	0.035*	0.023*	0.023*	0.023*	0.023*	0.426
	post	0.076	0.095	0.059	0.045*	0.035*	0.318
	second	0.185	0.026*	0.023*	0.026*	0.023*	0.185
Forehead: medial	pre	--	--	--	--	--	--
	first	0.089	0.010*	0.035*	0.010*	0.014*	0.185
	post	0.010*	0.010*	0.035*	0.010*	0.014*	0.652
	second	0.029*	0.010*	0.019*	0.010*	0.014*	0.215
Ear 1: dorsal	pre	--	--	--	--	--	--
	first	0.197	0.009**	0.016*	0.009**	0.013*	0.691
	post	0.009**	0.009**	0.016*	0.009**	0.013*	0.479
	second	0.126	0.009**	0.013*	0.009**	0.009**	1.000
Ear 2: medial	pre	--	--	--	--	--	--
	first	0.265	0.012*	0.016*	0.012*	0.012*	0.265
	post	0.070	0.010*	0.012*	0.010*	0.010*	0.820
	second	0.010*	0.010*	0.012*	0.010*	0.010*	0.155
Ear 5: medial	pre	--	--	--	--	--	--
	first	0.013*	0.013*	0.027*	0.013*	0.013*	0.479
	post	0.013*	0.066	0.066	0.013*	0.013*	0.778
	second	0.013*	0.013*	0.018*	0.013*	0.013*	0.820
Tail: wag amplitude	pre	--	--	--	--	--	--
	first	0.016*	0.131	1.000	0.016*	0.801	0.333
	post	--	--	--	--	--	--
	second	0.193	0.193	0.193	0.117	0.016*	0.783

Table 6: Post-hoc Wilcoxon signed-rank tests between pairs of different exposure phases, computed separately for each measurement and stimulus. We report post-hoc tests only for corresponding significant Friedman tests ($p_{FDR} \leq 0.05$, Table 4). – denotes no significant corresponding Friedman test. All reported p-values are Benjamini-Hochberg corrected within each measurement across all tested pairs. Note: All 40 pairwise comparisons between the social and food stimuli during the first and second exposure are highlighted in orange, as they show that we can separate between the food and social stimuli.

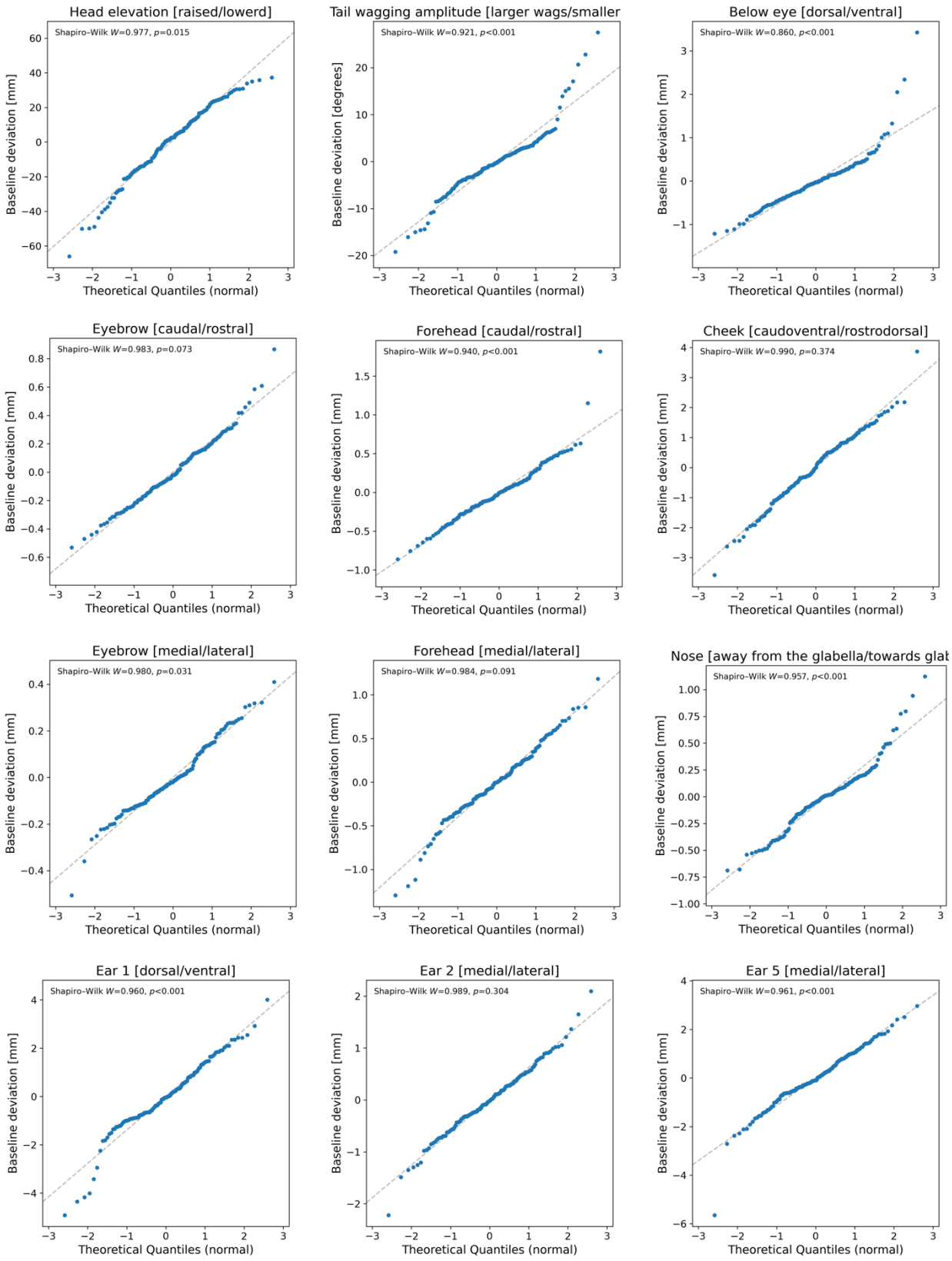


Table 7: Summary of normal distribution tests for all measurements.